

Unit Economics: CAC, LTV & Cross-Sell Propensity

Which acquisition channel is worth more, why customer lifetime value barely moves after the first purchase, which segments lose money once freight is netted out, and which product a customer is likely to buy next.

Core Question

A marketplace has a seller-acquisition funnel, a customer base, and a catalogue spanning dozens of categories and every region of the country. Which acquisition channel deserves more budget, which customer/product/region segments are quietly unprofitable, and can next-purchase behaviour be predicted well enough to run a cross-sell program?

Glossary

Term	Meaning
MQL	Marketing Qualified Lead - a prospective seller who entered the funnel via a tracked channel
Win rate	Share of MQLs that become a signed, live seller
Activation	A won seller who actually sold at least one item on the platform (winning is not the same as producing revenue)
GMV	Gross Merchandise Value - total item price sold, before any costs
Contribution margin (this project)	item price - freight value only - see docs/margin_proxy_definition.md for exactly what this excludes
Cohort	Customers grouped by the calendar month of their first purchase
Out-of-time split	Model trained on earlier months, tested on strictly later months - never a random split on time-ordered data
Propensity score	Predicted probability a customer adds a specific product next month, given they don't already have it

Key Findings

Finding	Value
Order items analysed	112,650 (98,666 orders), Sep 2016 - Oct 2018
Total GMV / freight-adjusted margin	R\$13.59M / 83.4% overall
Best acquisition channel (excl. "unknown")	paid_search - channel value score 0.71, beats organic_search (0.63)
Overall seller-funnel win rate	10.5% (842 of 8,000 MQLs)
Won sellers who never actually sold anything	55.0% (463 of 842) - winning ≠ activation
Customer repeat-purchase share	3.0% - 97.0% of customers buy exactly once
Cohort value growth, month 0 -> month 12	R\$111.41 -> R\$123.45 per customer (+10.8% , almost entirely from the 3% who return)
Worst-margin high-volume category	electronics - 31.5% margin, freight consumes 68.6% of price
Best-margin high-volume category	watches_gifts - 83.0% margin on R\$1.20M GMV
Worst-margin states	RO, RR (Amazon-region) - ~40% margin, freight >59% of price
Best-margin state	SP (São Paulo, where most sellers are based) - 73.5% margin
Cross-sell propensity model	16 of 24 products modelled (8 too rare to score, skipped)
Propensity model performance	Out-of-time test AUC 0.85 - 1.00 across products (median ~0.91)

Deliverables

- **Interactive dashboard:** `dashboards/p09_dashboard.html` - open directly in any browser, no server needed

- **Recommendation memo:** reports/01_RECOMMENDATION_MEMO.md
- **Executive summary:** reports/00_EXECUTIVE_SUMMARY.md
- **Charts:** outputs/charts/ - channel scorecard, funnel, cohort curves, repeat share, category/state margin, propensity AUC
- **Full source register + methodology notes:** reports/03_SOURCE_REGISTER.md, docs/
- **Engine:** src/*.py (each module runs standalone) + notebooks/ (same logic, walkthrough form)

Methodology

Two datasets, because no single public one carries both a multi-product repeat-customer panel and an acquisition funnel with channel and outcome data attached:

1. **Olist Brazilian E-Commerce + Marketing Funnel** (CC BY-NC-SA) - the seller-acquisition funnel (Part A), customer cohort economics (Part B), and segment margin analysis (Part C).
2. **Santander Product Recommendation panel** (1.5 years, ~950k customers, 24 products) - the cross-sell propensity model (Part D). Olist customers don't repeat-purchase often enough to model product-holding transitions.

Two things this analysis does not compute, both because the input isn't disclosed:

- **No dollar CAC.** Olist's marketing spend by channel isn't public anywhere, so channels are ranked on a composite of what is disclosed - win rate, activation rate, GMV per lead. See docs/seller_cac_methodology.md.
- **No payment fees or platform take-rate in margin.** "Contribution margin" here means price - freight, nothing else. See docs/margin_proxy_definition.md.

The propensity model trains on data through February 2016 and is evaluated on March-May 2016 (out-of-time split). Products with fewer than 30 training positives or 5 test positives were dropped rather than scored on too little signal - 8 of 24 fell out on that rule.

Data

- Olist Brazilian E-Commerce Public Dataset (olistbr/brazilian-ecommerce, CC BY-NC-SA 4.0)
- Olist Marketing Funnel (olistbr/marketing-funnel-olist, CC BY-NC-SA 4.0)
- Santander Product Recommendation dataset (public bank panel; original Kaggle competition data, mirrored as padmanabhanporaiyar/santander-product-recommendation-parquet-data)

Full provenance, license terms, and retrieval dates: reports/03_SOURCE_REGISTER.md.

Limitations & Assumptions

See reports/LIMITATIONS.md for the complete list. Headline items:

- Seller-side and customer-side analyses sit on the same platform but cover two separate populations (sellers vs. buyers). Nothing here models an interaction between them.
- The propensity model runs on a random 120,000-customer sample of the ~950,000-customer panel, for tractability. Sampling is uniform over customer IDs, but transition rates were not re-validated against the full panel.
- No payment-processing fee or platform commission is netted into any margin figure - see docs/margin_proxy_definition.md.

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License

Code: MIT (see LICENSE). The datasets keep their own licenses; terms and retrieval dates are in reports/03_SOURCE_REGISTER.md.